

LA SERENA, Chile

WMO: 854880

Lat: 29.9175S

Lon: 71.2011W

Elev: 142

StdP: 99.63

Time Zone: -4.00 (CHL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.6	6.3	1.2	4.2	8.8	3.0	4.8	8.4	6.8	13.6	6.3	13.6	2.5	120	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.3	22.7	17.2	21.8	16.8	21.0	16.5	17.7	21.4	17.2	20.7	16.8	20.2	5.3	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	11.7	19.4	15.7	11.4	18.9	15.2	11.0	18.4	50.6	21.6	49.1	20.9	47.7	20.3	20.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.0	6.4	6.0	3.5	24.6	1.3	1.5	2.6	25.6	1.9	26.5	1.2	27.3	0.3	28.3
			2.0	18.6	1.2	0.9	1.1	19.2	0.4	19.8	-0.3	20.3	-1.2	20.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.3	17.7	17.8	16.6	14.6	13.1	11.7	11.2	11.8	12.5	13.5	14.8	16.2
	DBStd	2.64	1.18	1.24	1.41	1.62	1.61	1.61	1.56	1.41	1.29	1.12	1.03	1.19
	HDD10.0	9	0	0	0	0	0	3	5	2	0	0	0	0
	HDD18.3	1500	26	22	57	113	163	198	220	203	175	151	107	66
	CDD10.0	1570	240	219	204	137	96	55	43	57	75	108	143	193
	CDD18.3	20	8	8	3	0	0	0	0	0	0	0	0	1
	CDH23.3	12	6	4	1	0	0	1	0	0	0	0	0	1
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	2.6	2.9	2.8	2.5	2.4	2.3	2.3	2.4	2.5	2.6	2.8	2.9	3.0
(15) Precipitation	PrecAvg	78	0	0	1	1	8	22	24	15	4	2	0	0
	PrecMax	223	2	2	14	26	69	159	177	96	31	21	5	10
	PrecMin	4	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	62	0	0	2	4	15	32	39	22	8	5	1	1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	24.1	22.9	21.0	20.1	18.7	18.6	19.1	19.1	19.8	20.6	22.2
		MCWB	17.9	18.3	17.2	15.7	14.7	13.5	12.9	13.2	13.8	13.7	15.0	16.0
	2%	DB	22.6	22.7	21.4	20.0	18.4	17.1	16.8	17.1	17.5	18.4	19.5	21.0
		MCWB	17.3	17.4	16.6	15.4	14.1	12.9	12.4	12.7	13.0	13.5	14.5	16.0
	5%	DB	21.8	21.8	20.7	19.0	17.2	16.0	15.5	16.0	16.6	17.6	18.8	20.1
		MCWB	17.0	17.0	16.3	14.8	13.5	12.6	11.8	12.2	12.5	13.2	14.2	15.6
	10%	DB	20.9	20.9	19.8	17.9	16.1	15.0	14.6	15.0	15.8	16.8	17.9	19.2
		MCWB	16.6	16.7	16.0	14.5	13.1	12.1	11.5	11.8	12.2	12.9	13.9	15.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.9	18.7	18.0	16.7	15.7	15.0	14.0	14.4	14.6	14.8	15.7	17.1
		MCDB	22.7	23.0	21.5	19.5	18.4	17.4	16.8	17.3	17.6	18.1	18.9	20.3
	2%	WB	17.7	17.8	17.2	16.0	14.8	13.8	13.1	13.4	13.7	14.1	15.1	16.4
		MCDB	21.4	21.6	20.5	18.7	17.4	15.9	15.6	15.9	16.6	17.4	18.4	19.7
	5%	WB	17.2	17.2	16.6	15.4	14.1	13.1	12.4	12.8	13.1	13.6	14.6	15.9
		MCDB	20.9	20.8	19.7	18.0	16.4	15.2	14.8	15.2	15.8	16.7	17.7	19.2
	10%	WB	16.7	16.9	16.1	14.9	13.5	12.5	11.8	12.1	12.5	13.1	14.1	15.5
		MCDB	20.2	20.2	19.1	17.4	15.7	14.5	14.0	14.5	15.0	16.0	17.1	18.7
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	6.3	6.2	6.3	6.3	6.8	7.2	6.9	6.7	6.7	6.4	6.3
		MCDBR	7.2	7.2	7.2	7.6	7.7	8.2	8.5	8.7	8.4	8.3	7.8	7.2
		MCWBR	3.4	3.5	3.7	4.3	4.7	5.4	5.6	5.6	5.0	4.6	4.1	3.6
	5% WB	MCDBR	6.8	6.6	6.6	6.7	7.2	7.2	7.8	7.8	7.4	7.4	6.9	6.7
		MCWBR	3.4	3.3	3.5	4.0	4.6	5.0	5.3	5.3	4.7	4.3	3.8	3.5
(40) Clear-Sky Solar Irradiance	taub	0.379	0.368	0.354	0.342	0.325	0.308	0.307	0.317	0.336	0.333	0.339	0.354	
	taud	2.399	2.426	2.457	2.478	2.512	2.566	2.555	2.511	2.440	2.452	2.460	2.439	
	Ebn at Noon	962	954	934	894	863	859	876	909	938	978	994	989	
	Edh at Noon	127	120	110	97	84	75	79	92	109	116	119	123	
(44) All-Sky Solar Radiation	RadAvg	7.83	7.25	6.02	4.55	3.23	2.81	3.07	3.89	5.14	6.52	7.50	7.99	
(45)	RadStd	0.38	0.30	0.38	0.36	0.29	0.27	0.23	0.32	0.32	0.45	0.39	0.40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.71	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page